

Homework 1a - Distractor Generation

Group 6

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Abstract

This document, will describe the methodology used in order to solve the problem of distractor generation within Task 26. This task consists in a classification task, for which we are going to generate a list of words that are closer in contextual meaning to the right label.

1 Introduction

The input dataset is structured into **three main file types**: vocabulary files, data files, and gold files, each with a specific role in defining and organizing terms and their hyperonymic relationships for hypernym discovery tasks. This translates to where, given a term (concept or entity) as input, our model has to retrieve a list with the most appropriate hypernyms from a text corpus.

The main goal of this project is to manipulate this dataset in order to produce as output a classification task, including the generation of **distractors**. Distractors are items or terms that are introduced in a recognition task to test the model's ability to discriminate between the correct answer and other plausible but incorrect options.

2 Technique

Initially, various approaches were tested to find distractors, including WordNet, Sense2Vec, and fine-tuning the models. However, all of them led to unsatisfactory results due to the fact that most language models are trained using English vocabularies.

The final technique implemented to generate the distractors involved a combination of Fasttext library and cosine similarity approach. Additionally, a new Italian vocabulary was utilized instead of the one provided by the task. This allowed us to achieve some viable distractors and to enhance the overall results, despite the sparsity of our vocabulary.

Due to time constraints and lack of hardware resources, we decide to work only with 30 terms from the training data. By *randomly* selecting the input terms, we aim to generalize the results across the entire dataset.

2.1 Tools

First, we download the **Italian database** containing 280,000 Italian words [1]. Then, we downloaded **FastText** [2], a library that can be used for efficient learning of word representations and sentence classification. Specifically, we employ the model "cc.it.300.bin" [3], which is a pre-trained model on Italian-language corpus.

2.2 Cosine Similiarity

With the new database of vocabulary and the Fast-text library, we computed the *embedding* of all the dataset. Then, using **cosine similarity**, we compared these embeddings with the embedding of the target word in order to find words that are most similar to the target word. The cosine similarity returns a value between 0 and 1, indicating the *degree of similarity* between the two words.

2.3 Filtering Function

We initially set a threshold of 0.5. For each candidate distractor, if it crosses the threshold, we compute its cosine similarity with the target word. However, this approach also resulted in words containing the target word (e.g., "stanza" -> "stanzina", "stanzuccia"). To avoid these *ambiguous words*, we implemented a **filtering function** that checks if the candidate distractor shares more than 40% of its letters with the target word. If it does, it will be discarded; otherwise, it will be selected as one of the three possible distractors.

After observing that some words did not return any distractor in output using the chosen threshold, we decided to decrease it by 0.025 and repeat the procedure until we reach the appropriate number

of distractors in output.

2.4 Final Step

Once we have a sufficient number of candidate distractors, we randomly select 3 of them in order to increase the *variance* of the results.

3 Instruction to run the code

To run the code from Google Colab, it is necessary to download the original dataset and upload it on your own Google Drive. Running the code, all the tools will be automatically downloaded.

If you are running the code from your own PC, you need to download FastText and the pre-trained model for Italian language from the provided link [3] and clone the repository from the link [1] to have the word dataset.

4 Metrics of the distractors word embeddings

From metrics in Figure 1, we can observe that there is a high variety in the choice of distractors but moderate semantic similarity, due to the fact that we always have to find at least three different distractors for every term and some terms have a slightly uncommon semantic meaning to be capable of finding highly related distractors.

Listing 1: Distractor Metrics

```
Metrics for training data:
Average Semantic Similarity: 0.194
Average Lexical Diversity: 0.899
```

```
Metrics for test data:
Average Semantic Similarity: 0.193
Average Lexical Diversity: 0.883
```

5 Conclusion

To conclude, this report describes how the integration of Fasttext, cosine similarity and new vocabulary could result into the generation of distractors. These distractors could be integrated with the classification task in order to be used in a generative language model.

References

- [1] Napolux. Italian words list. <https://github.com/napolux/paroleitaliane>.
- [2] Facebook AI Research. Fasttext documentation: Crawl vectors. <https://fasttext.cc/docs/en/crawl-vectors.html>.
- [3] Facebook AI Research. Word vectors for 157 languages. <https://fasttext.cc/docs/en/crawl-vectors.html>.